

Boss Challenge — Paper 12 (Class 7)

Nutrition in Animals | Soil | Data Handling | Exponents & Powers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Animals cannot make their own food and so depend on other living things; such organisms are called:
(A) decomposers
(B) heterotrophs
(C) autotrophs
(D) producers
2. The loose, top layer of the Earth's surface in which plants grow is called:
(A) clay
(B) soil
(C) humus
(D) bedrock
3. To work out the mean of several readings, the correct step is to:
(A) picking the largest number
(B) multiplying all the numbers
(C) divide the sum of the readings by how many there are
(D) adding the numbers only
4. In the expression $2x$, the number 2 is called the:
(A) coefficient
(B) exponent
(C) base
(D) product
5. The breaking down of complex food into simpler forms that the body can absorb is called:
(A) ingestion
(B) absorption
(C) digestion
(D) egestion
6. The slow breaking down of rocks into tiny particles that helps form soil is called:
(A) evaporation
(B) percolation
(C) erosion
(D) weathering
7. The mean of the numbers 4, 6 and 8 is:
(A) 6
(B) 3
(C) 8
(D) 18

8. In the expression 2^5 , the small raised number 5 is called the exponent or:
- (A) base
 - (B) sum
 - (C) power
 - (D) root
9. The teeth used mainly for biting and cutting food at the front of the mouth are the:
- (A) canines
 - (B) premolars
 - (C) incisors
 - (D) molars
10. A cut section showing the different layers of soil from the surface down is called the:
- (A) humus layer
 - (B) soil texture
 - (C) soil profile
 - (D) water table
11. The value that appears most often in a data set is called the:
- (A) mode
 - (B) range
 - (C) mean
 - (D) median
12. The value of 2^5 is:
- (A) 16
 - (B) 10
 - (C) 25
 - (D) 32
13. The pointed teeth on each side, used for tearing and piercing food, are the:
- (A) canines
 - (B) wisdom teeth
 - (C) incisors
 - (D) molars
14. The dark, rotted remains of dead plants and animals that make the topsoil fertile is called:
- (A) humus
 - (B) sand
 - (C) gravel
 - (D) clay
15. The mode of the data 2, 3, 3, 5, 3, 7 is:
- (A) 7
 - (B) 3
 - (C) 5
 - (D) 2

16. Using the law of exponents, $2^3 \times 2^9$ equals:

- (A) 2^{11}
- (B) 4^9
- (C) 2^9
- (D) 2^{12}

17. The watery liquid mixed with food in the mouth that begins to digest starch is called:

- (A) plasma
- (B) bile
- (C) mucus
- (D) saliva

18. The uppermost soil layer, dark and rich in humus where most roots grow, is called the:

- (A) bedrock
- (B) parent rock
- (C) topsoil
- (D) subsoil

19. When data values are arranged in order, the middle value is called the:

- (A) range
- (B) mode
- (C) median
- (D) mean

20. Using the law of exponents, $5^3 / 5^2$ equals:

- (A) 5^5
- (B) 5^3
- (C) 5^1
- (D) 1^5

21. Food is pushed down from the mouth to the stomach through a muscular tube called the:

- (A) trachea
- (B) oesophagus (food pipe)
- (C) rectum
- (D) small intestine

22. Soil that has the largest particles, lets water drain very fast and feels gritty is:

- (A) sandy soil
- (B) clayey soil
- (C) humus
- (D) loamy soil

23. The median of the ordered data 3, 5, 7, 9, 11 is:

- (A) 11
- (B) 5
- (C) 9
- (D) 7

24. Using the law of exponents, $(3^2)^3$ equals:

- (A) 3■
- (B) 9^3
- (C) 3■
- (D) 3■

25. The muscular movements of the food-pipe wall that push food forward are called:

- (A) filtration
- (B) respiration
- (C) circulation
- (D) peristalsis

26. Soil made of the smallest particles, which holds a lot of water and feels sticky when wet, is:

- (A) gravel
- (B) clayey soil
- (C) sandy soil
- (D) loamy soil

27. The difference between the highest and the lowest value in a data set is called the:

- (A) range
- (B) mean
- (C) mode
- (D) median

28. Any non-zero number raised to the power zero, such as 7^0 , equals:

- (A) 1
- (B) 7
- (C) 70
- (D) 0

29. The acid released in the stomach that kills germs and helps digest protein is:

- (A) carbonic acid
- (B) citric acid
- (C) sulphuric acid
- (D) hydrochloric acid

30. The soil considered best for growing most crops, a balanced mixture of sand, clay and humus, is:

- (A) rocky soil
- (B) clayey soil
- (C) sandy soil
- (D) loamy soil

31. For the data 12, 7, 20, 5, 16 the range is:

- (A) 5
- (B) 20
- (C) 15
- (D) 25

32. The value of 10^5 is:

- (A) 1000
- (B) 40
- (C) 100000
- (D) 10000

33. Most of the absorption of digested food into the blood takes place in the:

- (A) small intestine
- (B) liver
- (C) large intestine
- (D) stomach

34. The downward movement of water through the soil is measured as the rate of:

- (A) evaporation
- (B) weathering
- (C) percolation
- (D) erosion

35. A graph that shows information using pictures or symbols to stand for a number is called a:

- (A) pie chart
- (B) pictograph
- (C) bar graph
- (D) line graph

36. In standard form (scientific notation), the number 4500 is written as:

- (A) 4.5×10^2
- (B) 4.5×10^3
- (C) 0.45×10^5
- (D) 45×10^2

37. The tiny finger-like projections in the small intestine that increase the surface for absorption are the:

- (A) alveoli
- (B) nephrons
- (C) cilia
- (D) villi

38. Among sandy, clayey and loamy soils, the highest rate of water percolation is shown by:
- (A) loamy soil
 - (B) clayey soil
 - (C) sandy soil
 - (D) all are equal
39. A graph that uses rectangular bars of equal width to compare quantities is a:
- (A) histogram of curves
 - (B) bar graph
 - (C) pie chart
 - (D) pictograph
40. Written in standard form, the number 60000 is:
- (A) 60×10^3
 - (B) 6×10^3
 - (C) 6×10^4
 - (D) 6×10^5
41. Bile, which helps in the digestion of fats, is made by the:
- (A) liver
 - (B) kidney
 - (C) stomach
 - (D) pancreas
42. The removal of the fertile topsoil by wind or flowing water is called soil:
- (A) erosion
 - (B) pollution
 - (C) percolation
 - (D) weathering
43. Marks are recorded as |||| (four strokes) with the fifth stroke crossing them. These marks are called:
- (A) tally marks
 - (B) decimal points
 - (C) Roman numerals
 - (D) bar marks
44. The number 3.2×10^2 written as an ordinary number is:
- (A) 3.2
 - (B) 3200
 - (C) 320
 - (D) 32

45. Water from the undigested food is mainly absorbed back into the body in the:
- (A) mouth
 - (B) large intestine
 - (C) oesophagus
 - (D) small intestine
46. Planting trees and grass on bare land mainly helps the soil by:
- (A) holding the soil and preventing erosion
 - (B) turning sand into clay
 - (C) removing all humus
 - (D) making the soil drain faster
47. A graph comparing two sets of data side by side using paired bars is called a:
- (A) double bar graph
 - (B) pictograph
 - (C) single bar graph
 - (D) pie chart
48. The value of $(-2)^3$ is:
- (A) 6
 - (B) -8
 - (C) 8
 - (D) -6
49. Removal of the undigested, solid waste from the body through the anus is called:
- (A) excretion
 - (B) digestion
 - (C) egestion
 - (D) absorption
50. Adding too much fertiliser, plastic and harmful chemicals to the land causes soil:
- (A) erosion
 - (B) weathering
 - (C) percolation
 - (D) pollution
51. When a coin is tossed once, the probability of getting a head is:
- (A) 2
 - (B) 0
 - (C) $\frac{1}{2}$
 - (D) 1
52. The value of $(-2)^4$ is:
- (A) 16
 - (B) 8
 - (C) -8
 - (D) -16

53. Grass-eating animals like the cow that quickly swallow grass and later chew the cud are called:
- (A) carnivores
 - (B) omnivores
 - (C) ruminants
 - (D) decomposers
54. Which type of soil is most suitable for growing paddy (rice), which needs standing water?
- (A) rocky soil
 - (B) loamy soil
 - (C) clayey soil
 - (D) sandy soil
55. When an ordinary dice is rolled once, the probability of getting the number 4 is:
- (A) $\frac{4}{6}$
 - (B) $\frac{1}{6}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{1}{4}$
56. Which of these means 'three multiplied by itself five times'?
- (A) $3■$
 - (B) 5^3
 - (C) $3 + 5$
 - (D) 3×5
57. In ruminants, the swallowed grass that is returned to the mouth to be chewed again is called the:
- (A) saliva
 - (B) cud
 - (C) bile
 - (D) rennet
58. The soil layer just below the topsoil, harder and containing little humus, is called the:
- (A) humus
 - (B) bedrock
 - (C) topsoil
 - (D) subsoil
59. In a pictograph, one symbol stands for 5 books. To show 35 books, the number of symbols drawn is:
- (A) 30
 - (B) 35
 - (C) 7
 - (D) 5

60. Expressed as a power of 2, the number 16 is:
- (A) 2■
 - (B) 2■
 - (C) 4^2
 - (D) 2^3
61. Tiny one-celled Amoeba captures its food using finger-like extensions called:
- (A) cilia
 - (B) villi
 - (C) tentacles
 - (D) pseudopodia
62. The hard, solid rock found at the bottom of the soil profile is called the:
- (A) topsoil
 - (B) bedrock
 - (C) humus
 - (D) loam
63. Over five days a plant grew 2, 4, 3, 5 and 6 cm. Its average daily growth was:
- (A) 6 cm
 - (B) 4 cm
 - (C) 20 cm
 - (D) 5 cm
64. A soil sample is found to contain about 1000000 tiny organisms. In standard form this count is:
- (A) $1 \times 10^{\text{■}}$
 - (B) $1 \times 10^{\text{■}}$
 - (C) $10 \times 10^{\text{■}}$
 - (D) $1 \times 10^{\text{■}}$
65. Inside Amoeba, the food it has trapped is digested within a:
- (A) food vacuole
 - (B) gizzard
 - (C) rumen
 - (D) stomach
66. Earthworms and tiny burrowing animals help the soil mainly by:
- (A) removing all its water
 - (B) turning it into bedrock
 - (C) making it loose and airy so water and air pass through
 - (D) making it harder and packed

67. A weather record shows rainfall (mm) on four days as 10, 20, 15 and 15. The mean daily rainfall is:
- (A) 15 mm
 - (B) 20 mm
 - (C) 4 mm
 - (D) 60 mm
68. Using exponent laws, $10^4 \times 10^3$ equals:
- (A) 10^7
 - (B) 10^2
 - (C) 10^1
 - (D) 100^7
69. The broad, flat back teeth used for grinding and chewing food are the:
- (A) molars
 - (B) canines
 - (C) milk teeth
 - (D) incisors
70. The amount of water that a soil can hold after the extra has drained away is called its water:
- (A) retention
 - (B) erosion
 - (C) percolation
 - (D) evaporation
71. The leaves on six plants were counted as 5, 8, 8, 8, 10 and 6. The most common (mode) number of leaves is:
- (A) 8
 - (B) 5
 - (C) 6
 - (D) 10
72. The reading 7.0×10^3 grams is the same mass as:
- (A) 7000 grams
 - (B) 70 grams
 - (C) 70000 grams
 - (D) 700 grams
73. The first set of teeth that appear in young children and later fall out are called:
- (A) permanent teeth
 - (B) canine teeth
 - (C) wisdom teeth
 - (D) milk teeth

74. Clayey soil holds more water than sandy soil mainly because clay has:
- (A) more stones in it
 - (B) no gaps at all
 - (C) larger particles with wide gaps
 - (D) smaller particles with tinier gaps that trap water
75. To compare the number of trees planted by boys and by girls in each of four years, the best graph is a:
- (A) line of best fit
 - (B) single bar graph
 - (C) pictograph for one group
 - (D) double bar graph
76. Which number is the largest?
- (A) 4^2
 - (B) 2^4
 - (C) 5^1
 - (D) 3^2
77. The muscular organ in the mouth that mixes food with saliva and helps in tasting is the:
- (A) epiglottis
 - (B) tongue
 - (C) gall bladder
 - (D) uvula
78. Plants depend on soil mainly to provide them with anchorage and with:
- (A) ready-made food
 - (B) light for photosynthesis
 - (C) oxygen for the leaves
 - (D) water and minerals
79. If the mean of three numbers is 10, then the sum of the three numbers is:
- (A) 30
 - (B) 10
 - (C) 13
 - (D) 3
80. The value of 1 raised to any power, such as 1^{100} , is:
- (A) 0
 - (B) 10
 - (C) 1
 - (D) 100

81. Animals such as snakes that feed only on other animals are called:
- (A) herbivores
 - (B) omnivores
 - (C) carnivores
 - (D) autotrophs
82. Which property of soil decides how easily roots can push through and breathe in it?
- (A) its colour
 - (B) its name
 - (C) its smell
 - (D) how loose and airy (well-aerated) the soil is
83. The chance of an event that is certain to happen (like the Sun rising) is written as a probability of:
- (A) 1
 - (B) 100
 - (C) $\frac{1}{2}$
 - (D) 0
84. Written as a single power, $6^3 \times 6^2 \times 6$ equals:
- (A) 6^6
 - (B) 6^6 written as 18^3
 - (C) 6^6
 - (D) 6^6
85. Bile released by the liver is stored, before use, in a small sac called the:
- (A) spleen
 - (B) pancreas
 - (C) appendix
 - (D) gall bladder
86. A simple way to judge a soil's texture (sandy, clayey or loamy) is to:
- (A) weigh it on a balance
 - (B) feel and rub a moist sample between the fingers
 - (C) smell it from far
 - (D) look at its colour only
87. In 30 tosses of a coin, heads came up 18 times. The number of times tails came up is:
- (A) 18
 - (B) 48
 - (C) 30
 - (D) 12

88. The distance is given as 9.3×10^7 km. The exponent 7 tells you that the first part is multiplied by:
- (A) 7
 - (B) 10 multiplied by itself 7 times
 - (C) 70
 - (D) 10 added 7 times
89. An animal like a human that eats both plants and other animals is called an:
- (A) herbivore
 - (B) omnivore
 - (C) carnivore
 - (D) scavenger
90. The constant cover of growing plants protects soil chiefly because their leaves and roots:
- (A) make the soil drain faster
 - (B) slow the rain and bind the soil, reducing erosion
 - (C) add stones to the soil
 - (D) stop weathering of rocks
91. A pictograph shows one tree-symbol for every 100 trees and there are 6 full symbols. The number of trees shown is:
- (A) 60
 - (B) 6
 - (C) 106
 - (D) 600
92. Expressed as a power of 10, the number 1000 is:
- (A) 10^2
 - (B) 10^4
 - (C) 3×10^3
 - (D) 10^3
93. The process of taking food into the body through the mouth is called:
- (A) digestion
 - (B) absorption
 - (C) ingestion
 - (D) assimilation
94. If equal water is poured into a sandy and a clayey sample, the sandy sample will:
- (A) turn the water into vapour
 - (B) let the water drain out faster
 - (C) hold the water longer
 - (D) not let any water pass

95. The mean mass of four soil samples is 50 g. If three of them are 40 g, 55 g and 45 g, the fourth sample's mass is:

- (A) 50 g
- (B) 60 g
- (C) 140 g
- (D) 200 g

96. The value of 3^3 is:

- (A) 6
- (B) 27
- (C) 9
- (D) 33

97. The opening at the end of the digestive tract through which faeces leave the body is the:

- (A) mouth
- (B) oesophagus
- (C) nostril
- (D) anus

98. In a percolation test, 200 mL of water drained through a soil sample in 20 minutes. The percolation rate is:

- (A) 4000 mL per minute
- (B) 20 mL per minute
- (C) 10 mL per minute
- (D) 200 mL per minute

99. On a bar graph of rainfall in five cities, the city whose bar is the tallest is the one with the:

- (A) least rainfall
- (B) greatest rainfall
- (C) no rainfall at all
- (D) average rainfall

100. Written in standard form, the number 250000 is:

- (A) 2.5×10^5
- (B) 2.5×10^6
- (C) 25×10^4
- (D) 2.5×10^4